

AI/ML intern DAY 11

TASK — Dataset Collection and Analysis for Virtual Try-On Systems

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Introduction

The objective of this task was to collect and study datasets used in Artificial Intelligence-based Virtual Try-On Systems, Human Pose Estimation, Body Segmentation, and Fashion Recommendation technologies.

The research focused on:

- fashion clothing datasets
- human pose datasets
- body segmentation datasets
- apparel try-on datasets

The task helped understand how AI models are trained for:

- virtual fitting systems
- garment alignment
- pose estimation
- segmentation workflows
- intelligent fashion recommendation

Dataset Areas Covered

Fashion Clothing Images

Studied datasets containing:

- clothing images
- garment categories
- outfit styles
- fashion attributes

These datasets help train:

- fashion classification systems
 - recommendation engines
 - virtual fitting workflows
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Human Pose Datasets

Studied datasets used for:

- body landmark detection
- skeletal tracking
- motion analysis
- posture estimation

These datasets improve:

- AI pose estimation
 - body alignment
 - motion tracking systems
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Body Segmentation Datasets

Studied datasets focused on:

- human parsing
- body masking
- foreground extraction
- clothing segmentation

These datasets are used for:

- virtual try-on systems
 - cloth overlay workflows
 - body-cloth separation
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Apparel Try-On Datasets

Studied datasets designed for:

- virtual garment fitting
- clothing transfer

- outfit generation
- AI try-on systems

These datasets help train:

- garment alignment models
- virtual fitting engines
- fashion AI systems

Datasets Collected

Dataset Name	Source	Use Case
DeepFashion	Kaggle	Fashion image analysis
VITON Dataset	GitHub/Kaggle	Virtual try-on systems
Fashion-MNIST	Zalando Research	Clothing classification
CIHP Dataset	Human Parsing Research	Human segmentation
ATR Dataset	Human Parsing Research	Cloth segmentation

1. DeepFashion Dataset

Purpose

DeepFashion is a large-scale fashion dataset used for:

- fashion recognition
 - clothing analysis
 - outfit recommendation
 - fashion classification
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Features

- clothing categories
 - fashion attributes
 - clothing landmarks
 - garment images
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Applications

- AI fashion recommendation
- virtual fitting systems
- fashion analytics
- clothing classification



Figure 2. Example images of different categories and attributes in DeepFashion. The attributes form five groups: texture, fabric, shape, part, and style.

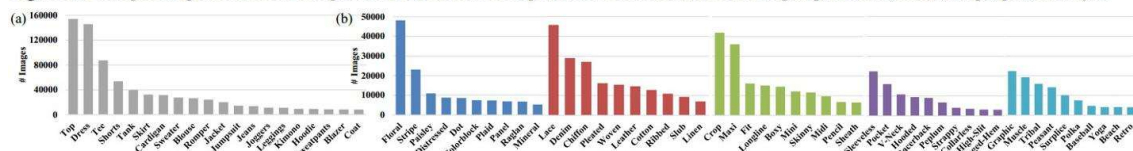


Figure 3. (a) Image number of the top-20 categories. (b) Image number of the top-10 attributes in each group.

Figure 1 — DeepFashion Dataset Workflow

2. VITON Dataset

Purpose

VITON is a dataset designed specifically for:

- virtual try-on systems
- garment transfer
- clothing overlay workflows

Features

- person images
- clothing images
- paired garment data

- try-on samples

Applications

- AI clothing overlay
 - virtual fitting systems
 - apparel transfer models
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Figure 2 — VITON Virtual Try-On Dataset

3. Fashion-MNIST Dataset

Purpose

Fashion-MNIST is a dataset used for:

- clothing classification
- machine learning training
- AI fashion prediction

Features

- shirt images
 - footwear images
 - clothing labels
 - grayscale fashion images
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Applications

- AI classification systems
 - machine learning training
 - clothing prediction workflows
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Figure 3 — Fashion-MNIST Dataset

4. CIHP Dataset

Purpose

CIHP is a human parsing dataset used for:

- body segmentation
- human parsing
- body masking

Features

- body region labels
- segmentation masks
- body part extraction

Applications

- body segmentation
- virtual fitting systems
- human parsing workflows



(a) MPII and LSP



(b) ATR and PASCAL-Person-Part



(c) LIP

5. ATR Dataset

Purpose

ATR Dataset is used for:

- cloth segmentation
- body parsing
- garment extraction

Features

- human parsing labels
- clothing segmentation
- body-cloth separation

Applications

- AI segmentation systems
- cloth extraction workflows
- garment fitting systems



Dataset Collection Workflow

Fashion Dataset Collection



Human Pose Datasets



Body Segmentation Datasets



Virtual Try-On Datasets



AI Model Training



Virtual Fitting System

Technologies Used

- Artificial Intelligence
 - Machine Learning
 - Computer Vision
 - Deep Learning
 - Human Parsing Systems
 - Fashion Recommendation Systems
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Expected Outputs

- dataset research documentation
 - dataset workflow analysis
 - AI model training research
 - segmentation dataset analysis
 - fashion dataset comparison
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Research Findings

The research helped understand:

- AI dataset preparation
- body segmentation training
- fashion classification workflows
- garment alignment systems
- virtual fitting model training

The datasets provide training data for:

- virtual try-on systems
- pose estimation
- cloth segmentation

- AI fashion recommendation
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Learning Outcomes

During this task, the following concepts and skills were learned:

- AI dataset collection
 - fashion image analysis
 - body segmentation datasets
 - pose estimation datasets
 - virtual try-on model training
 - garment classification workflows
 - AI data preprocessing
 - computer vision applications
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Conclusion

The Dataset Collection Task successfully analysed multiple Artificial Intelligence datasets used in Virtual Try-On Systems, Human Parsing, Fashion Recommendation, and Body Segmentation workflows.

The task improved understanding of:

- AI fashion datasets
- pose estimation datasets
- body segmentation workflows
- virtual try-on model training
- AI recommendation systems
- garment classification
- computer vision applications

This task focused on Artificial Intelligence, Machine Learning, Computer Vision, Virtual Try-On Systems, Human Parsing, Fashion Recommendation, and AI Dataset Analysis Workflows.